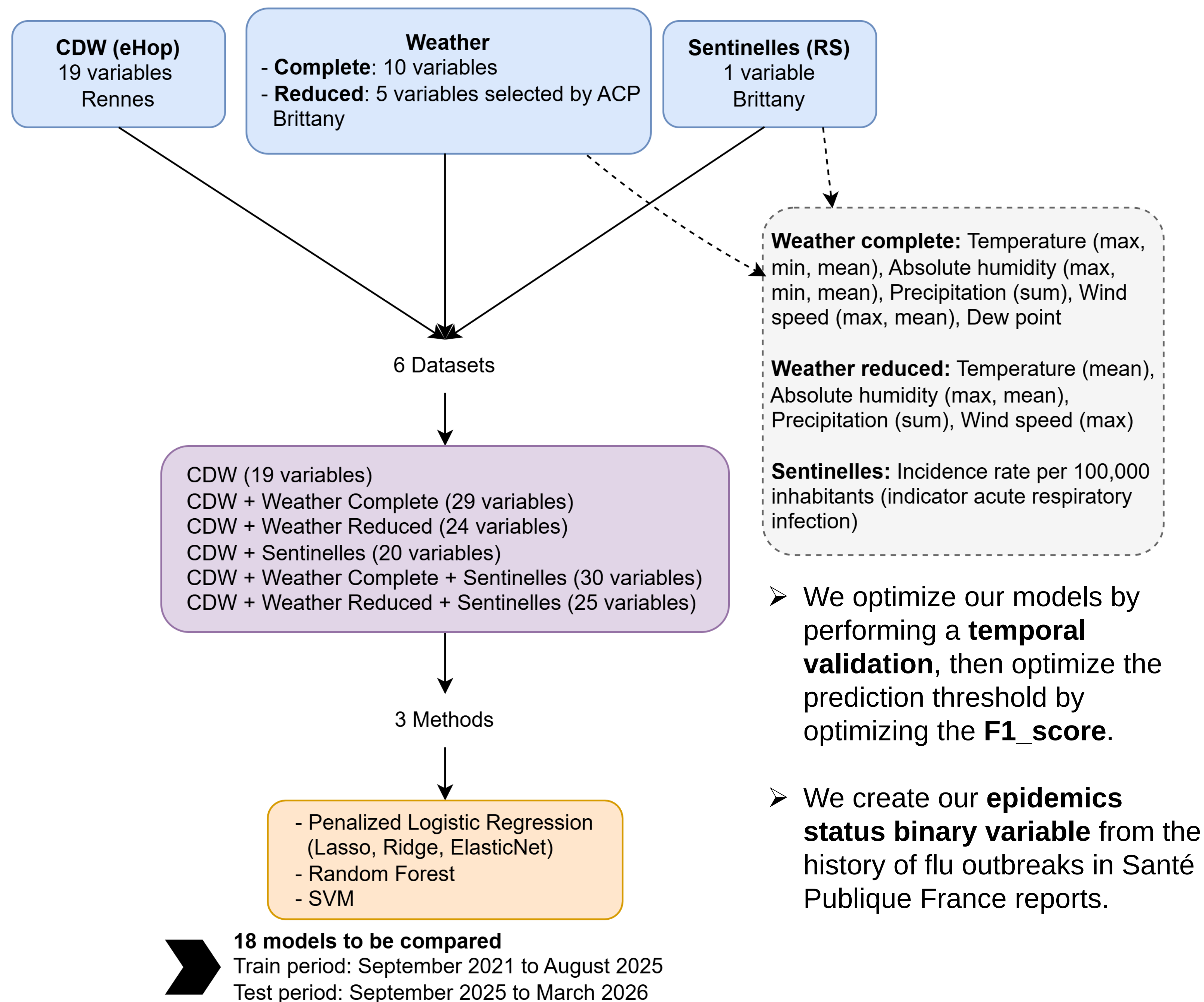


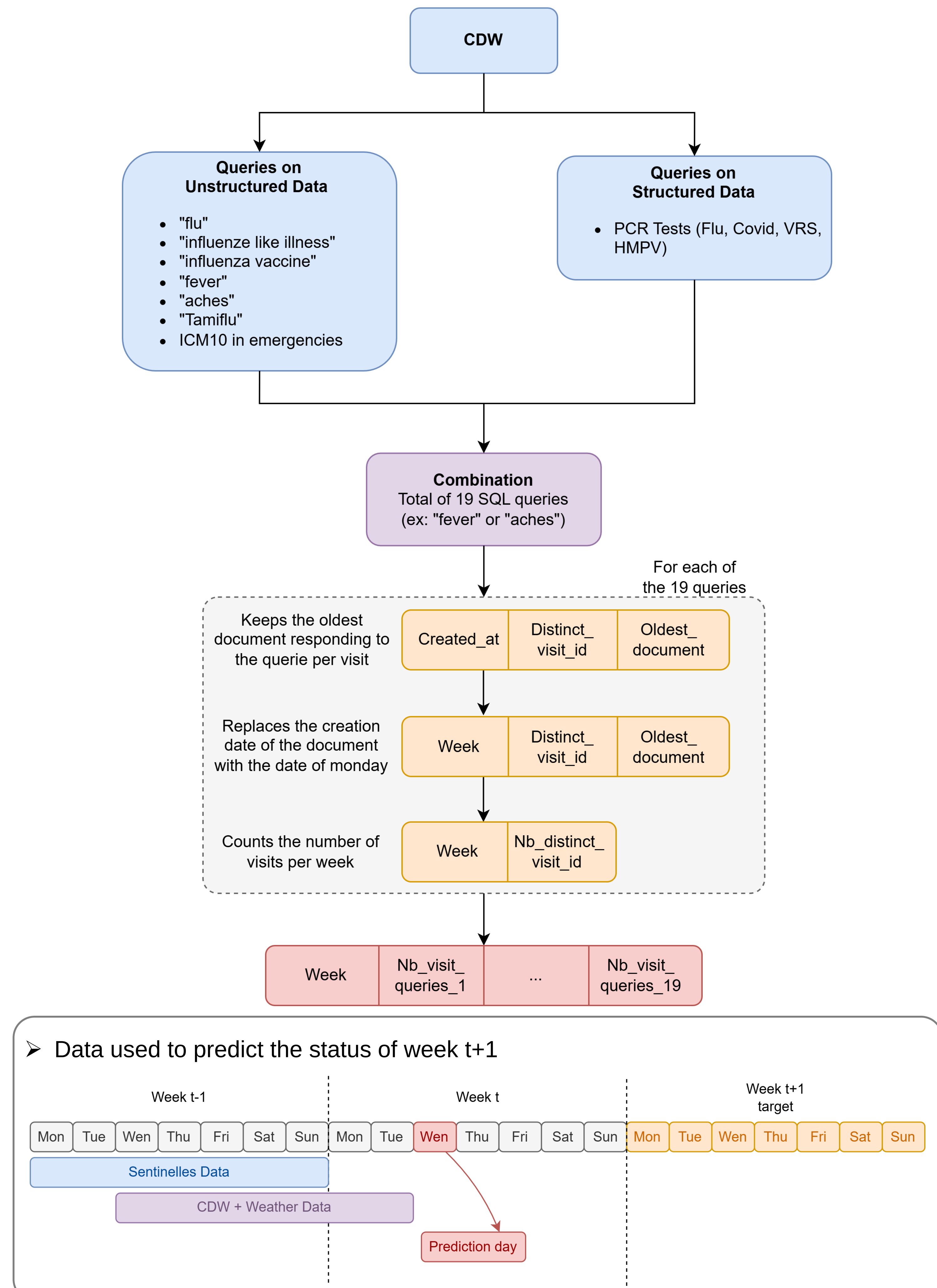
## Introduction

- Each year, one billion people are affected by **influenza** worldwide, causing between **290,000 and 650,000 deaths**.
- In France, traditional flu surveillance relies on the **Sentinelles system**, which provides valuable but delayed indicators (1-3 weeks).
- **Clinical Data Warehouses (CDWs)** can reduce this lag by using structured and unstructured hospital data to deliver timely, validated case counts enriched with clinical details.
- The objective is to improve responsiveness to outbreaks by exploiting CDWs combined with heterogeneous sources, in order to **predict the epidemic status** of the following week, with a particular emphasis on **early detection of the outbreak**.

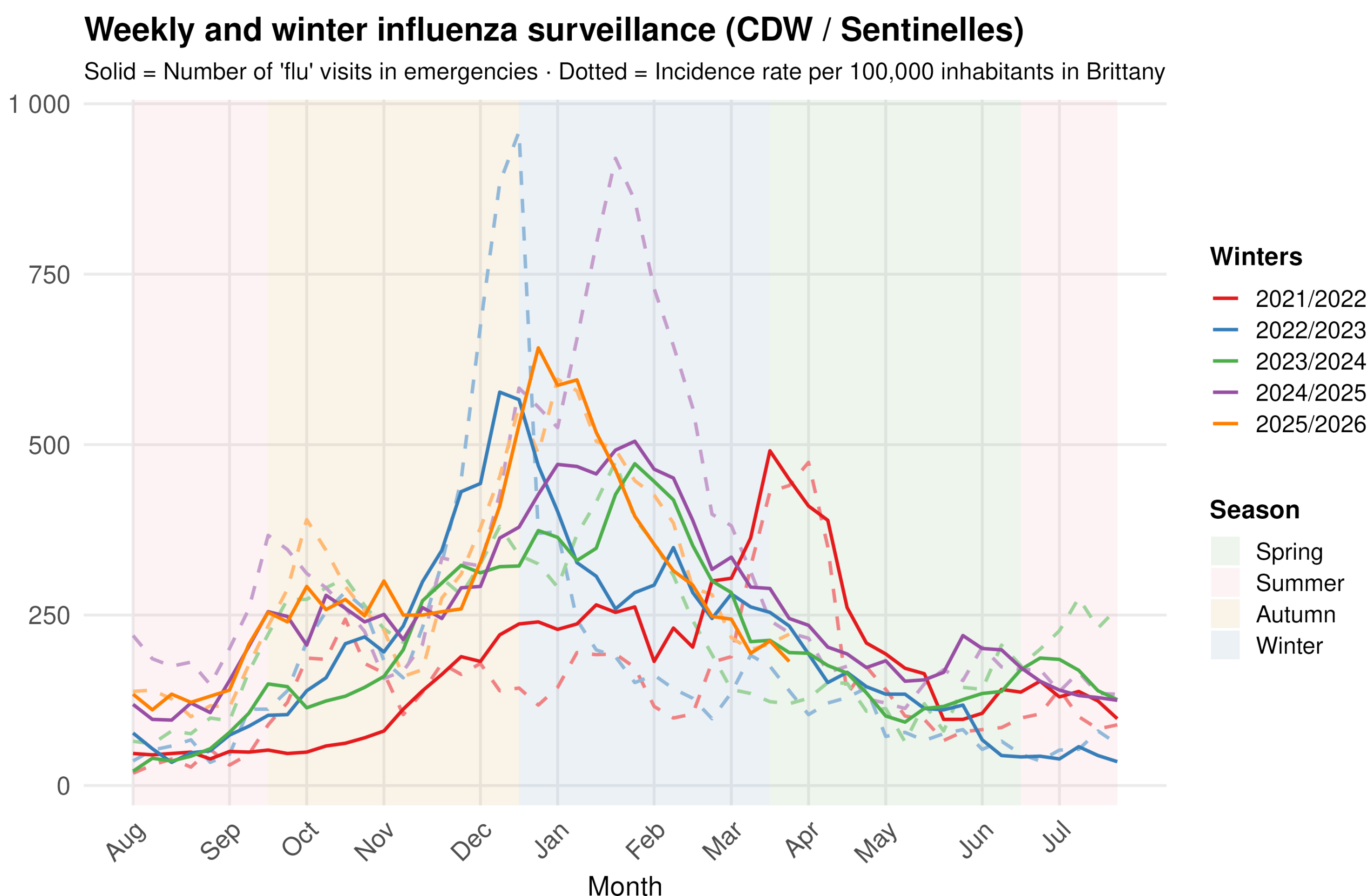
## 2. Combining data extraction and predicted models



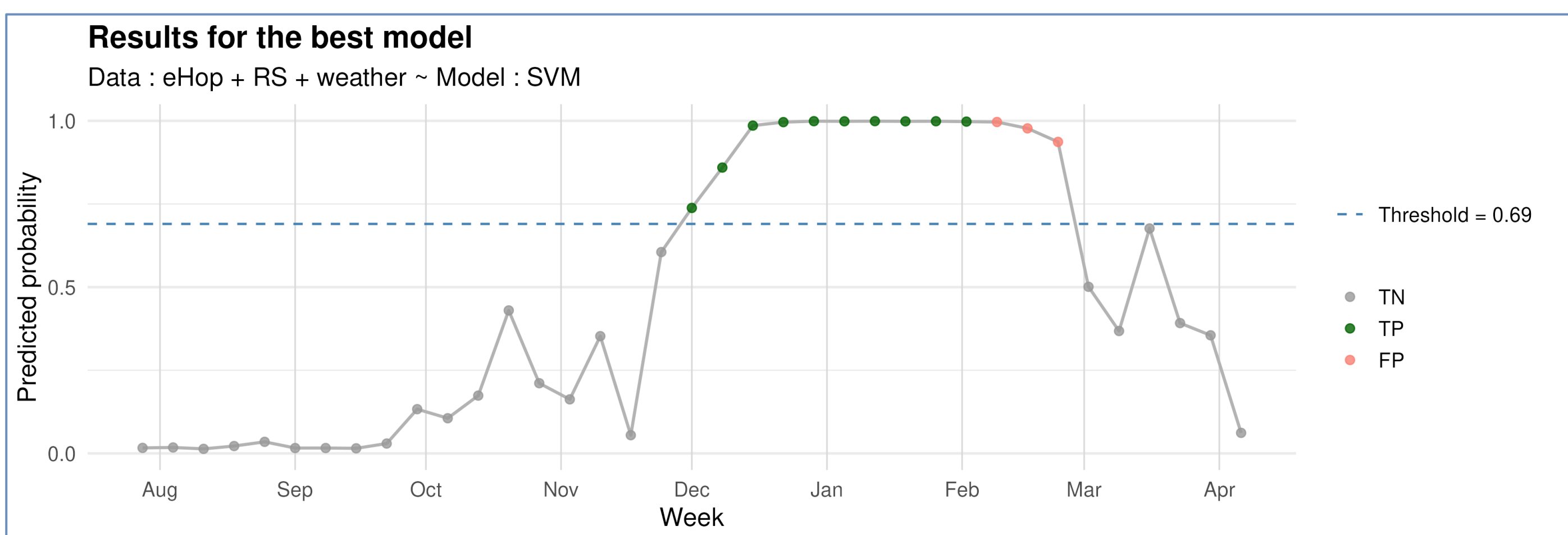
## 1. Data extraction



## Results



| Model                   | TP | FP | TN | FN | Recall | Precision | F1    | DeltaStart | DurationDiff |
|-------------------------|----|----|----|----|--------|-----------|-------|------------|--------------|
| rl_ehop                 | 8  | 2  | 25 | 2  | 0.8    | 0.800     | 0.800 | 2          | 0            |
| rl_ehop_weather_red     | 8  | 2  | 25 | 2  | 0.8    | 0.800     | 0.800 | 2          | 0            |
| rf_ehop                 | 8  | 1  | 26 | 2  | 0.8    | 0.889     | 0.842 | 2          | -1           |
| svm_ehop                | 8  | 1  | 26 | 2  | 0.8    | 0.889     | 0.842 | 2          | -1           |
| svm_ehop_weather_com    | 10 | 4  | 23 | 0  | 1.0    | 0.714     | 0.833 | 0          | 3            |
| svm_ehop_weather_red    | 10 | 4  | 23 | 0  | 1.0    | 0.714     | 0.833 | 0          | 3            |
| svm_ehop_weather_com_RS | 10 | 3  | 24 | 0  | 1.0    | 0.769     | 0.870 | 0          | 3            |
| svm_ehop_weather_red_RS | 10 | 4  | 23 | 0  | 1.0    | 0.714     | 0.833 | 0          | 3            |



## Conclusion

- We propose **combining heterogeneous data sources** through machine learning models to consistently **improve forecasting accuracy** and **responsiveness** of epidemiological indicators.
- It is possible to get closer to real time thanks to the **CDW**, whose data are **updated with a delay of day+1**. It is even possible to go further by trying to predict the beginning of an epidemic the following week.
- Future work could explore deep learning approaches such as **MLPs** and perform **weight tuning** to optimize the contribution of each variable and improve prediction performance.

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GitHub repository with the poster and the authors' contacts